Fiscal Year:	FY 2026	**Task Last Updated:**	FY 07/14/2026
PI Name:	Zubair, Abba M.D., Ph.D.		
Project Title:	Role of Mesenchymal Stem Cells in Microgravity Induced Bone Loss		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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City:	32224-1865	**State:**	FL
Zip Code:	32224-1865	**Congressional District:**	4
Comments:			
Project Type:		**Solicitation / Funding Source:**	03-OBPR-02
Start Date:	02/01/2021	**End Date:**	01/31/2027
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 01/31/2027 per F. Hernandez/ARC (Ed., 3/10/26) NOTE: End date changed to 01/31/2026 per NSSC information (Ed., 12/17/24) NOTE: End date changed to 01/31/2025 per F. Hernandez/ARC (Ed., 6/18/24) NOTE: Period of performance updated to 2/1/21-1/31/24 (from 12/1/20-11/30/23) per F. Hernandez/ARC and NSSC information (Ed., 2/2/21)		
Key Personnel Changes/Previous PI:	DECEMBER 2022 Dr. Mannie Saunders is no longer collaborating in this research project because she has retired from research and closed her lab at University of Akron.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K0299		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bone loss resulting in low bone mineral density as observed in osteopenia and osteoporosis increases the risk of fracture. Bone loss is a major problem at Earth gravity (1 g) and is aggravated by microgravity in space, especially during long duration space flight. Exercise slows bone loss in microgravity; but even when astronauts adopt a rigorous exercise regimen, they continue to lose about 1 – 2% of their bone mass a month. While there is a vast body of knowledge on bone homeostasis under normal gravity conditions, the mechanisms of bone loss in microgravity are poorly understood and remain to be explored. It is not yet fully determined what signals allow bone tissue to adapt to a weightless environment compared to an Earth environment. Conceptually, altered biomechanical stimuli in microgravity may directly affect osteoblasts, osteoclasts, and/or their precursors, while other physiological factors such as hormone levels or poor nutrition could contribute to bone loss in microgravity. The central hypothesis of our proposal is that microgravity alters the activity of mesenchymal stem cells, which are the precursors of bone forming osteoblasts, and their secretion of cytokines and growth factors that promote osteogenesis and osteolytic activity. Specifically, we postulate that microgravity may decrease the mesenchymal stem cells (MSCs) number and disturb the secretion of trophic factors that impede bone homeostasis by reducing new bone formation and/or increasing bone resorption in microgravity. Bone homeostasis requires a complex interplay between mesenchymal osteoblasts and hematopoietic osteoclasts and is controlled by multiple cytokines, including the pro-osteolytic factors interleukin (IL)-1 and tumor necrosis factor a (TNFa), and the anti-osteolytic factors transforming growth factor ß (TGFß), IL-4, and IL-10 that are pro-osteogenesis. Some cytokines such as IL-6 pleiotropically affect bone homeostasis. Our preliminary results show that microgravity culturing of MSCs at the International Space Station (ISS) does not significantly alter MSC proliferation, but alters their cytokine and growth factor secretion with an associated increase in the secretion of pro-osteolytic cytokines. We and others have shown that IL-6 secretion by MSC is very critical in MSC functioning. We propose that microgravity skews MSC secretomes towards osteolysis and that IL-6 plays a central role in the perturbation of the homeostatic balance. IL-6 activity is very critical in bone homeostasis, but IL-6 effects depend on how its signal is transmitted within its target cell. IL-6 acts via the classic signaling pathway by binding to its membrane bound receptor (IL-6R) and homodimeric glycoprotein 130 (gp130) proteins. In addition, IL-6 can act via its soluble receptor (sIL-6) coupled with gp130 protein to act on non-IL-6R expressing target cells as part of a non-canonical trans-signaling pathway. IL-6 induces osteogenesis and anti-osteolytic activities through its classical signaling pathway and induces osteolysis via its trans-signaling pathway. We postulate that microgravity provokes MSC induced bone loss and suppresses new bone formation via activation of the IL-6 trans-signaling pathway and blockade of the classic pathway. The rationale for our proposed studies is that a biological understanding of the IL-6 signaling pathways in MSC induced bone loss may permit pharmacological activation or blockage of the classical and trans-signaling pathways as a new and innovative approach to management of bone loss both in space and on Earth. Our long-term goal is to understand the mechanism of MSC induced bone loss under microgravity conditions and to develop safe and effective therapy for bone loss. The principal experimental objectives are to determine how IL-6 signaling pathways contribute to MSC induced bone loss under microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The rationale for our proposed studies is that a biological understanding of IL-6 signaling pathways and its downstream effects in MSC induced bone loss, may permit pharmacological activation or blockage of the classic and trans-signaling pathways as a new and innovative approach to management of bone loss both in space and on Earth. Our long-term goal is to understand the mechanism of MSC induced bone loss under microgravity conditions and to develop a safe and effective therapy for bone loss.
Task Progress:	Bone loss is a serious issue for people on Earth and becomes even worse for astronauts in space, especially during long space missions. Exercise can slow down bone loss in space, but astronauts still lose about 1 to 2% of their bone mass each month, even if they exercise regularly. Our research aims to better understand how a molecule called IL-6 affects bone loss by studying its signals in special cells called mesenchymal stem cells (MSCs). If we know how these signals work, we might be able to create new medicines to help prevent bone loss, both in space and on Earth. To study this, we planned two experiments, MABL-A and MABL-B which involved sending samples to space and doing similar tests in our lab at the Mayo Clinic to mimic space conditions. The MABL-A experiment was sent to the International Space Station on January 30, 2024, and came back on April 30, 2024. We studied samples from space, from our lab on Earth, and from special equipment that copies space conditions. Our main focus was on how MSC cells act differently in these environments. Right now, we are writing a report about what we discovered. For our second experiment, called MABL-B, we made sure everything would work by doing a test earlier this year. We sent the experiment to space on SpaceX-33 on August 24, 2025. Astronauts on the International Space Station took care of the samples by changing their growth liquid and collecting them when ready. At the same time, we did the same steps here on Earth in our lab. The samples from space are expected to come back in February 2026. In summary, we have finished looking at most of the results from our first experiment, MABL-A, and are writing two papers to share what we learned. We plan to send these papers to science journals early next year. The work for MABL-B in space and on Earth is also done, and once we get the samples back from space, we will study them and write a paper about our findings.
Bibliography Type:	Description: (Last Updated: 07/14/2026)
Abstracts for Journals and Proceedings	Zubair A. "Stem cells, microgravity and longevity." WIRED Summit 2025: The Big Interview, Mexico City, Mexico, September 18, 2025. Abstracts. WIRED Summit 2025: The Big Interview, Mexico City, Mexico, September 18, 2025. , Sep-2025
Abstracts for Journals and Proceedings	Zubair A. "From Down Under to the stars: Advancements and opportunities in human health in space." International Astronautical Congress 2025, Sydney, Australia, September 29-October 3, 2025. Abstracts. International Astronautical Congress 2025, Sydney, Australia, September 29-October 3, 2025. , Sep-2025

Abstracts for Journals and Proceedings	Zubair A. "Microgravity associated bone loss: Role of mesenchymal stem cells." International Space Station National Lab Prelaunch Science Webinar: SpX-33, Virtual, August 19. 2025. Abstracts. International Space Station National Lab Prelaunch Science Webinar: SpX-33, Virtual, August 19. 2025. , Aug-2025
Significant Media Coverage	Ciepielewski CGS. (Zubair A interview). "He is Abba Zubair, the researcher who is growing stem cells in microgravity. Dr. Zubair interviewed for WIRED en Español." WIRED Magazine, November 23, 2025. https://es.wired.com/articulos/platicamos-con-el-investigador-que-esta-cultivando-celulas-madre-en-microgravedad , Nov-2025